

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: TV Tuner
Manufacturer: Contemporary Research
Firmware Version: N/A
Model(s): ATSC-SDI 4i

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.14.0000-b007	2.13.0

Version History

Module Version	Date	Notes
1_0_0_0	9/23/2021	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- DeviceID variable must be set accordingly. Default value is '1'. DeviceID ranges from '1' to '9', and Broadcast. If Device ID is set to 'Broadcast', status is unavailable.
Example: `InterfaceName.DeviceID = '1'`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='ATSC-SDI 4i')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='ATSC-SDI 4i')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 23, Model='ATSC-SDI 4i')</code>

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command ChannelDiscreteCommand ¹	Value 'String'		
# ChannelDiscreteCommand example InterfaceName.Set('ChannelDiscreteCommand', '1-1')			
Command ChannelStep	Value 'Up'	Value 'Down'	
# ChannelStep example InterfaceName.Set('ChannelStep', 'Up')			
Command ClosedCaption	Value 'On'	Value 'Off'	
# ClosedCaption example InterfaceName.Set('ClosedCaption', 'On')			
Command ClosedCaptionAnalog	Value 'Caption 1' 'Caption 4' 'Text 3'	Value 'Caption 2' 'Text 1' 'Text 4'	Value 'Caption 3' 'Text 2'
# ClosedCaptionAnalog example InterfaceName.Set('ClosedCaptionAnalog', 'Caption 1')			
Command ClosedCaptionDigital	Value 'Service 1' 'Service 4'	Value 'Service 2' 'Service 5'	Value 'Service 3' 'Service 6'
# ClosedCaptionDigital example InterfaceName.Set('ClosedCaptionDigital', 'Service 1')			
Command ExecutiveMode	Value 'Off' 'Ch+Vol+Menu' 'Menu' 'Power+Setup+Menu'	Value 'Ch+Menu' 'Power' 'All'	Value 'Vol+Menu' 'Setup' 'Setup+Menu'
# ExecutiveMode example InterfaceName.Set('ExecutiveMode', 'Off')			
Command Keypad	Value '0' – '9'	Value '_'	
# Keypad example InterfaceName.Set('Keypad', '0')			
Command MenuNavigation	Value 'Up' 'Right' 'Enter'	Value 'Down' 'Menu' 'Exit'	Value 'Left' 'Guide'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Up')			

Global Scripter Module Communication Sheet

Command Power	Value 'On'	Value 'Off'
# Power example InterfaceName.Set('Power', 'On')		
Command Volume	Value 0 to 100 in steps of 1	
# Volume example InterfaceName.Set('Volume', 100)		

¹ Channel Discrete String must be set to use Channel Discrete Command.

- For analog and one-part digital channels, the valid value range is 0 – 9999.
- For two-part digital channels, the valid value range is 0 – 999 for both the major and minor channel. Separate with major and minor channel with a dash (-).

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus, AudioMute, and Volume do not support the Update function. ConnectionStatus is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Value	Value	Value
AudioMute	'On'	'Off'	
# AudioMute example Value = InterfaceName.ReadStatus('AudioMute') InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command	Value	Value	Value
ClosedCaption ²	'On'	'Off'	
# ClosedCaption example InterfaceName.Update('ClosedCaption') Value = InterfaceName.ReadStatus('ClosedCaption') InterfaceName.SubscribeStatus('ClosedCaption', None, FeedbackHandler)			
Command	Value	Value	Value
ClosedCaptionAnalog	'Caption 1'	'Caption 2'	'Caption 3'
	'Caption 4'	'Text 1'	'Text 2'
	'Text 3'	'Text 4'	
# ClosedCaptionAnalog example InterfaceName.Update('ClosedCaptionAnalog') Value = InterfaceName.ReadStatus('ClosedCaptionAnalog') InterfaceName.SubscribeStatus('ClosedCaptionAnalog', None, FeedbackHandler)			
Command	Value	Value	Value
ClosedCaptionDigital	'Service 1'	'Service 2'	'Service 3'
	'Service 4'	'Service 5'	'Service 6'
# ClosedCaptionDigital example InterfaceName.Update('ClosedCaptionDigital') Value = InterfaceName.ReadStatus('ClosedCaptionDigital') InterfaceName.SubscribeStatus('ClosedCaptionDigital', None, FeedbackHandler)			
Command	Value	Value	
ConnectionStatus	'Connected'	'Disconnected'	
# ConnectionStatus example Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command	Value	Value	
Power ¹	'On'	'Off'	
# Power example InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)			

Command	Value
Volume	0 to 100 in steps of 1
<pre># Volume example Value = InterfaceName.ReadStatus('Volume') InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)</pre>	

¹This command also updates AudioMute and Volume statuses.

²This command also updates ClosedCaptionAnalog and ClosedCaptionDigital.

Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 9600

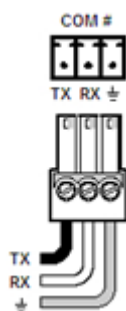
Data Bits: 8

Parity: None

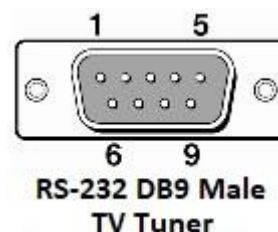
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD	→	2	RxD
RxD	←	3	TxD
GND		5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface.

Port Type: Ethernet (TCP)

Default Port: 23

Logon Credentials No

Supported:

Multi-Connection Undetermined

Capabilities:

Port Changeability: Yes

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Audio Mute Off	>1VX\x0D
Audio Mute On	>1VM\x0D
Channel Step Down	>1TD\x0D
Channel Step Up	>1TU\x0D
Closed Caption Analog Caption 1	>1Q1=1\x0D
Closed Caption Analog Caption 2	>1Q1=2\x0D
Closed Caption Analog Caption 3	>1Q1=3\x0D
Closed Caption Analog Caption 4	>1Q1=4\x0D
Closed Caption Analog Text 1	>1Q1=5\x0D
Closed Caption Analog Text 2	>1Q1=6\x0D
Closed Caption Analog Text 3	>1Q1=7\x0D
Closed Caption Analog Text 4	>1Q1=8\x0D
Closed Caption Digital Service 1	>1Q7=1\x0D
Closed Caption Digital Service 2	>1Q7=2\x0D
Closed Caption Digital Service 3	>1Q7=3\x0D
Closed Caption Digital Service 4	>1Q7=4\x0D
Closed Caption Digital Service 5	>1Q7=5\x0D
Closed Caption Digital Service 6	>1Q7=6\x0D
Closed Caption Off	>1Q0=0\x0D
Closed Caption On	>1Q0=1\x0D
Executive Mode All	>1S4=7\x0D
Executive Mode Ch+Menu	>1S4=1\x0D
Executive Mode Ch+Vol+Menu	>1S4=3\x0D
Executive Mode Menu	>1S4=6\x0D
Executive Mode Off	>1S4=0\x0D
Executive Mode Power	>1S4=4\x0D
Executive Mode Power+Setup+Menu	>1S4=9\x0D
Executive Mode Setup	>1S4=5\x0D
Executive Mode Setup+Menu	>1S4=8\x0D
Executive Mode Vol+Menu	>1S4=2\x0D
Keypad -	>1KK=99\x0D
Keypad 0	>1KK=10\x0D
Keypad 9	>1KK=19\x0D
Menu Navigation Down	>1KK=109\x0D
Menu Navigation Enter	>1KK=110\x0D
Menu Navigation Exit	>1KK=111\x0D
Menu Navigation Guide	>1KK=63\x0D
Menu Navigation Left	>1KK=107\x0D
Menu Navigation Menu	>1KK=29\x0D
Menu Navigation Right	>1KK=106\x0D
Menu Navigation Up	>1KK=108\x0D
Power Off	>1P0\x0D

Global Scripter Module Communication Sheet

Power On	>1P1\x0D
Volume 0	>1VH=0\x0D
Volume 100	>1VH=100\x0D

Appendix B. Update Commands

Closed Caption	>1SQ\x0D
Power	>1SV\x0D